

Seasonal SDE using Function OLS

0.1 Empirical Econometric Functions

1. Cao model -

Daily temperature residual follows a k-lag auto-correlation process (Cao)

$$U_t = \sum_{i=1}^k \rho_i U_{t-i} + \sigma \xi_t \quad (1)$$

where

$$\sigma_t = \sigma_0 - \sigma_1 \left| \sin \left(\frac{\pi t}{365} + \varphi \right) \right|$$

and

$$\xi_t \sim \text{i.i.d. } \mathcal{N}(0, 1)$$

2. Campbell model -

Seasonal volatility is approximated by a Fourier series and cyclical volatility by a GARCH model. Temperature model becomes -

$$T_t = \text{Trend}_t + \text{Seasonal}_t + \sum_{l=1}^L \rho_{t-l} T_{t-l} + \sigma_t \varepsilon_t \quad ()$$

where

$$\begin{aligned} \text{Trend}_t &= \sum_{m=0}^M \beta_m t^m \quad \text{and} \quad \text{Seasonal}_t = \sum_{p=1}^P \left(\sigma_{c,p} \cos \left(\frac{2\pi p d(t)}{365} \right) + \sigma_{s,p} \sin \left(\frac{2\pi p d(t)}{365} \right) \right) \\ \sigma_t^2 &= \sum_{q=1}^Q \left(\gamma_{c,q} \cos \left(\frac{2\pi q d(t)}{365} \right) + \gamma_{s,q} \sin \left(\frac{2\pi q d(t)}{365} \right) \right) + \sum_{r=1}^R \alpha_r (\sigma_{t-r} - \varepsilon_{t-r})^2 + \sum_{s=1}^S \beta_s \sigma_{t-s}^2 \end{aligned} \quad (2.11)$$

where $\varepsilon_t \sim \text{i.i.d.}(0, 1)$ and $d(t)$ is a repeating function that cycles 1 to 365.

3. Alaton Model -

Mean reverting process with monthly varying volatility -

$$dT_t = \frac{dT_m}{dt} + a(T_m^t - T_t)dt + \sigma dW_t \quad (2)$$

where

$$T_m^t = A + Bt + C \sin(wt + \varphi)$$

and $w = \frac{2\pi}{365}$, and φ is the phase angle.

4. Benth Model -

CAR(p) model, which is a continuous AR(p) time series representation, where temperature can be modeled as -

$$T(t) = \Lambda(t) + X_1(t) \quad (3)$$

where $\Lambda(t)$ is a deterministic seasonal function representing average temperature and the stochasticity is modeled through $X(t)$, which is a mean reverting Ornstein-Uhlenbeck process $\in \mathbb{R}^p$ for $p \geq 1$:

$$dX(t) = AX(t)dt + e_p\sigma(t)dB(t) \quad (4)$$

where $B(t)$ is a Brownian motion, e_k is the k -th unit vector in $\mathbb{R}^p$, $k = 1, \dots, p$, $\sigma_t > 0$ is a real-valued and square-integrable function and A is a $p \times p$ matrix of the form:

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -\alpha_p & -\alpha_{p-1} & -\alpha_{p-2} & \cdots & -\alpha_1 \end{bmatrix}$$

with explicit solution for $X(t) = x \in \mathbb{R}^p$ -

$$X(S) = \exp(A(S-t))x + \int_t^S \exp(A(S-u))e_p\sigma(u)dB(u) \quad (5)$$

$s \geq t$

5. Cabrera Model -

Average temperature t -th day:

$$T_t = \frac{T_{\max_t} + T_{\min_t}}{2}$$

following an Ornstein-Uhlenbeck (OU) process:

$$dT_t = a(T_m - T_t)dt + \sigma_t dW_t$$

Seasonality with a time-dependent mean that accounts for this seasonality implies -

$$\frac{dT_t^m}{dt} = B + \omega \cdot C \cdot \cos(\omega t + \phi)$$

0.2 SDE -

$$dT_t = \left\{ \frac{dT_t^m}{dt} + a(T_t^m - T_t) \right\} dt + \sigma_t dW_t$$

Distribution of differences $\Delta T_t = T_t - T_{t-1}$ has mean and variance :

$$\overline{\Delta T_t} = \frac{1}{n} \sum_{t=1}^n \Delta T_t$$

$$\frac{1}{n} \sum_{t=1}^n (\Delta T_t - \overline{\Delta T_t})^2.$$

Calibration

Following Alaton as well as Bibby and Sorensen in a *martingale estimation method* for drift and diffusion coefficients. This efficient estimator can be obtained by finding the value of $\hat{a}_n$ such that:

$$G_n(a) = 0 = \sum_{i=1}^n \frac{\partial_a b(T_{i-1}; a)}{\sigma_{i-1}^2} (T_i - \mathbb{E}[T_i | T_{i-1}])$$

Discretized SDE based variance estimator -

$$\hat{\sigma}_u^2 = \frac{1}{N_u - 2} \sum_{j=0}^{N_u-1} (T_j - \hat{a}T_{j-1} - (1 - \hat{a})T_{j-1})^2$$

0.3 OLS to be fitted -

$$T_t^m = a_1 + a_2 t + a_3 \sin(\omega t) + a_4 \cos(\omega t)$$

where:

- T_t^m : Modeled temperature at time t
- ω : Angular frequency, typically set to $\frac{2\pi}{365}$ for daily data with an annual cycle

such that -

$$T_t^m = A + Bt + C \sin(\omega t + \phi)$$

- $A = a_1$, $B = a_2$, $C = \sqrt{a_3^2 + a_4^2}$
- $\phi = \arctan\left(\frac{a_4}{a_3}\right) - \pi$

0.4 Alternate - Jump Model

0.4.1 Parameter Calibration

Using rolling window estimation, the simulation was designed to be run in two dimensions. Spatial one - run on different locations. Temporal one - to capture changes in the parameters through time for each location. Simulations were run using 5 years initial period and then following iterations added one more

year to previous one. In each turn error estimates, HDD and CDD estimates were calculated. Results were compared with actual HDD and CDD values. A turn, on the other hand, consisted of 10000 runs. At the end of each run, average values of 10000 runs were calculated to estimate the parameters. with whole process repeated using initial 6,7,8,... years of data. Finally, parameters of the year that offered best estimates of the HDDs and CDDs were chosen for predictions.

Discretization Discretization was done by using Euler approximation as following:

$$T_{t+1} = T_t + T_{t+1}^m + T_{t+1}^m + b(T_{t-1}^m - T_t) + H_{t+1} + \gamma Y_{t+1} - \alpha(Q_{t+1} - Q_t) + Z_t + \beta(R_{t+1} - R_t) \quad (6)$$

where $H_{t+1} = \sqrt{\gamma_0 + \gamma_1 \bar{H}_t \epsilon_t}$, γ_0 and γ_1 are ARCH parameters and $\epsilon \sim N(0, 1)$.

Mean reversion Parameters of T^m were found by least squares method. To make this, $Y_t = a_1 + a_2 t + a_3 \sin(wt) + a_4 \cos(wt)$ was fitted to temperature data [?]. Then, parameters were obtained by $A = a_1, B = a_2, C = \sqrt{a_3^2 + a_4^2}$ and $\varphi = \arctan\left(\frac{a_4}{a_3}\right) - \pi$.

Mean reversion parameter b were estimated as:

$$\hat{b} = -\log\left(\frac{\sum_{i=1}^n |(T_{i-1} - T_{i-1}^m)(T_i - T_i^m)|}{\sum_{i=1}^n (T_{i-1} - T_{i-1}^m)^2}\right)$$

Jump component

Mean reversion parameters of jump parts were estimated as $\hat{\alpha} = 1$ and

$$\hat{\beta} = -\log\left(\frac{\sum_{i=1}^n x_{i-1} - x_i}{\sum_{i=1}^n |x_{i-1}|^2}\right)$$